

Module 5

Complex Engineering Problems and Complex Engineering Activities

Training on OBE and Accreditation

Board of Accreditation for Engineering and Technical Education (BAETE)

 The logo of Bangladesh Applied University of Science and Technology (BAUST). It features a green shield with a gear and a circuit board design inside. The text 'BAUST' is written across the shield. Below the shield, a banner reads 'DISCIPLINE KNOWLEDGE MORALITY' and the year '2015' is displayed.	Presented by Mafin Muntasir Rahman Assistant Professor Dept. of EEE, BAUST	Courtesy of A. S. M. A. Haseeb Chairman, Training Taskforce, BAETE	 The logo of the Board of Accreditation for Engineering and Technical Education (BAETE). It is a circular emblem with a blue background and white text. The outer ring reads 'BOARD OF ACCREDITATION FOR ENGINEERING AND TECHNICAL EDUCATION'. The inner part features a large white checkmark and the acronym 'BAETE'.
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Outline

1. Introduction

2. Complex Engineering Problem Solving

3. Complex Engineering Activities

4. Implementation and Examples

5. Summary

What Does Engineers Do?

What Does Engineers Do?

Engineers

**Solve
Complex Problems**

No obvious solution;
require abstract thinking
and originality in analysis to
formulate suitable models.

Technologists

**Solve
Broadly-defined Problems**

Can be solved by well-
proven analysis techniques.

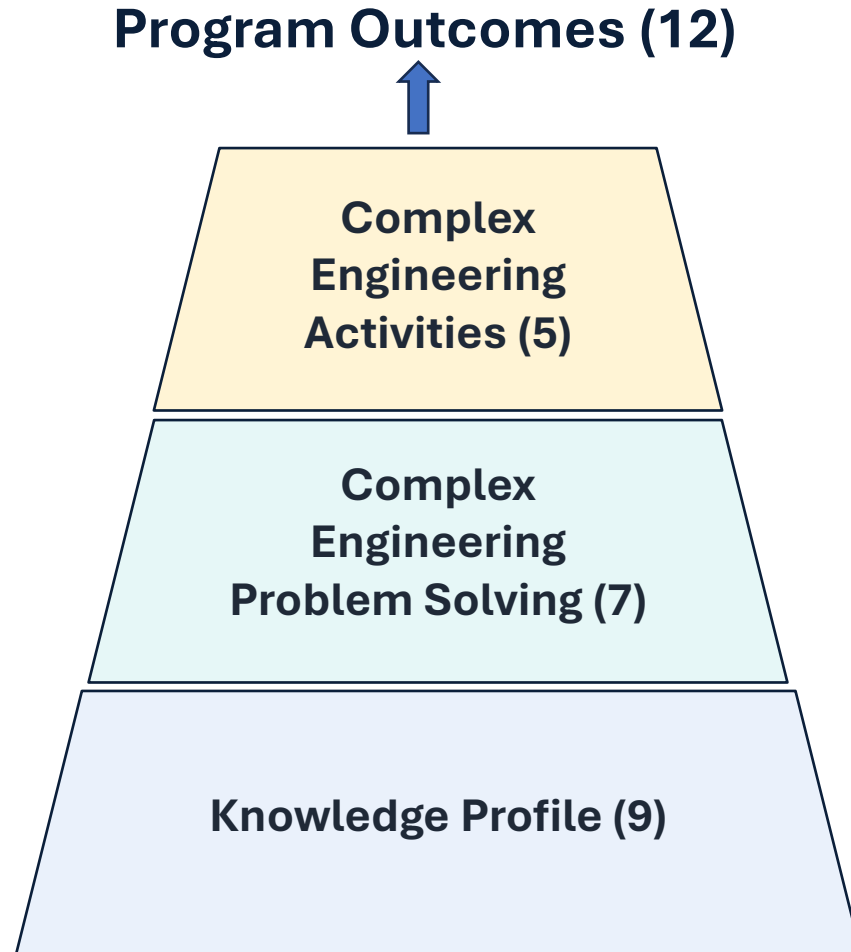
Technicians

**Solve
Well-defined Problems**

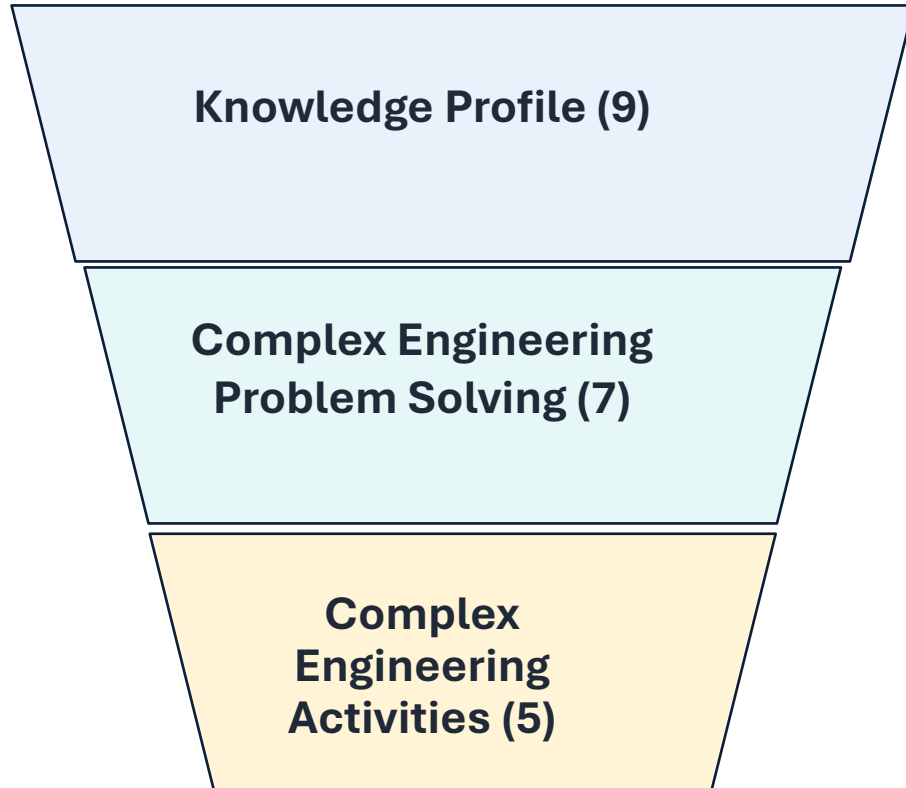
Can be solved in
standardized ways.

Program Outcomes and Engineering Complexity

Foundational knowledge must equip students to solve complex engineering problems and execute professional projects involving complex engineering activities.



Program Outcomes: How the Three Layers Connect



Program Outcomes (12)

Knowledge Profile

Math, science, engineering principles, social sciences, and other foundations a student must master.

Complex Engineering Problem Solving

Students use the knowledge profile to analyze abstract, open-ended, and non-standard engineering problems.

Complex Engineering Activities

Practical, project-based application requiring management of resources, team dynamics, and diverse stakeholders.

Ability to execute both CEP and CEA

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Attributes of Knowledge Profile (WK1–WK9)

WK	Attribute
WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences
WK2	Conceptually based mathematics, numerical analysis, data analysis, statistics and the formal aspects of computer and information science to support detailed analysis and modeling applicable to the discipline
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as professional responsibility of an engineer to public safety and sustainable development
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes

**Criteria for a problem to be
considered as a
Complex Engineering Problem?**

Range of Complex Engineering Problem Solving

Attribute	Complex Engineering Problems have characteristics WP1 and some or all of WP2 to WP7
Depth of knowledge required	WP1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamentals-based, first principles analytical approach
Range of conflicting requirements	WP2: Involve wide-ranging or conflicting technical, non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements
Depth of analysis required	WP3: Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models
Familiarity of issues	WP4: Involve infrequently encountered issues or novel problems
Extent of applicable codes	WP5: Address problems not encompassed by standards and codes of practice for professional engineering
Extent of stakeholder involvement and conflicting requirements	WP6: Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs
Interdependence	WP7: Address high level problems including many components or sub-problems that may require a systems approach

When Is It a Complex Engineering Problem?

Complex Engineering Problem = WP1 + at least two from WP2 to WP7

Knowledge Attributes Required for Solving CEP

WP1 requires

One or more
Knowledge
Attributes:

WK3
WK4
WK5
WK6
WK8

Code	Knowledge area
WK3	Engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge for accepted practice areas.
WK5	Resource use, environmental impacts, whole-life cost, re-use and net zero carbon.
WK6	Engineering practice/technology in the discipline.
WK8	Current research literature and emerging issues.

Can we differentiate between complex engineering problem and complicated engineering problem?

Complicated vs Complex Engineering Problems

Complicated Engineering Problem

- Predictable; follows strict rules, even with thousands of moving parts.
- Cause-and-effect relationships are linear and consistent.
- Governed by well-established formulas, standardized best practices.
- **Examples:**
 - Calculate structural steel requirements for a 10-story building using local zoning codes
 - Design a smartphone motherboard
 - Design a mechanical clock

Complex Engineering Problem

- Unpredictable; involves shifting variables with no single obvious solution.
- Cause-and-effect is non-linear and often outside standards or manuals.
- Solutions require abstract thinking and trade-offs between conflicting requirements.
- **Examples:**
 - Plan a municipal transportation network considering balanced highway alignment, environmental impact, economic profitability, and public health guideline across various interest groups.
 - Integrate renewable energy sources into a national smart power grid.

Discussion

Set up a 5-axis CNC machining center to manufacture an aerospace engine block.

Revitalize a coastal city's storm surge defense system to combat rising sea levels.

Design a secure and reliable campus-wide network using a fiber-optic backbone, managed switches, VLANs, and wireless access points to support a specified number of users, devices, and bandwidth requirements across multiple buildings.

Design a decentralized microgrid with consumer-owned solar panels, EV batteries, and localized wind turbines.

Design a high-efficiency turbocharger for a passenger car engine.

Design and manage a decentralized edge-computing network that securely processes data from distributed IoT devices while ensuring scalability, fault tolerance, and efficient resource allocation.

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Range of Complex Engineering Activities

Attribute	Complex activities mean (engineering) activities or projects that have some or all the following characteristics:
Range of resources	EA1: Involve the use of diverse resources including people, data and information, natural, financial and physical resources and appropriate technologies including analytical and/or design software
Level of interactions	EA2: Require optimal resolution of interactions between wide-ranging and/or conflicting technical, non-technical, and engineering issues
Innovation	EA3: Involve creative use of engineering principles, innovative solutions for a conscious purpose, and research-based knowledge
Consequences to society and the environment	EA4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
Familiarity	EA5: Can extend beyond previous experiences by applying principles-based approaches

Discussion

Question

Can a group work solving a problem of mechanics be considered as a Complex Engineering Activity?

Why? Why not?

Problem Set 1

1. Car and Bicycle Rider

A car is driving along a straight line with a speed v_0 . At time $t = 0$ the car is at the origin. At a later instant of time $t = t_1$ the car starts to slow down until it stops at a time $t = t_2$.

The acceleration of the car as a function of time is given by

$$a_c = \begin{cases} 0 & 0 < t < t_1 \\ -c(t - t_1) & t_1 < t < t_2 \end{cases}$$

where c is a positive constant which has dimensions of acceleration per unit time.

- Find $v_c(t)$ and $x_c(t)$, the x-component of the velocity and the position of the car as a function of time. Express your answer in terms of some or all of the following variables: v_0 , c , t , t_1 and t_2 .
- A bicycle rider is riding at a constant speed of v_b and at $t = 0$ is 17 m behind the car. The cyclist reaches the car when the car just comes to rest. The car is moving with an acceleration

$$a_c = \begin{cases} 0 & 0 < t < t_1 \\ -c(t - t_1) & t_1 < t < t_2 \end{cases}$$

where its initial component of the velocity is $v_0 = 12$ m/s, $t_1 = 1$ s and $c = 6$ m/s³. The car comes to rest at t_2 .

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How/When to Implement CEP and CEA?

- Accreditation bodies like BAETE do not provide a single prescription.
- Programs decide how to implement, but they need to provide evidence.

Evidence places

- **Theory courses**
- **Laboratory courses**
- **Design projects**
- **Final Year Design Project**

Mapping Example

Course	Program Outcomes												Knowledge Profile									Complex Engineering Problem Solving							Complex Engineering Activities				
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	WK1	WK2	WK3	WK4	WK5	WK6	WK7	WK8	WK9	WP1	WP2	WP3	WP4	WP5	WP6	WP7	WA1	WA2	WA3	WA4	WA5
	WK1-WK4	WK1-WK4	WK5	WK8	WK2 WK6	WK1 WK5 WK7		WK9			WK8																		Related to PO9				
	Needs Complex Engineering Problem Solution (WP and some of WP2 to WP7)																																
	Knowledge	Analysis	Design	Investigation	Tool Usage	Society & Environment	Ethics & Industry	Individual and Teamwork	Communication	Project Management and Finance	Life-long learning	Entrepreneurship	Science	Mathematics	Engineering Fundamentals	Engineering Specialization	Design	Technology	Society and Sustainability	Research and critical thinking	Ethics and Inclusivity	Knowledge WK3 - WK8	Wide Range/Conflicting	No Obvious Solution	Infrequent Issues	Problem Outside Codes	Diverse Stakeholders	Many Components	Range of Resources	Level of Interactions	Innovation	Consequences	Familiarity
Course 1	X			X			X					X	X	X	X				X	X	X		X		X								
Course 2		X				X	X					X	X	X	X	X		X		X	X			X	X								
Course 3			X					X								X												X			X		
Course 4					X				X				X				X					X	X			X							
Course 5	X	X					X					X	X	X	X					X	X		X		X								
Course 6	X					X		X				X	X	X	X	X		X		X	X		X		X								
Course 7					X								X				X				X		X		X								
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
FYDP		X	X	X	X	X	X	X	X	X	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Some examples from practices in different universities

Example 1: An In-class Assignment

- In this individual assignment, each student designs the fittings, fixtures, conduit layout (including cable types and sizes), switch board and distribution board connections (including rating of protective equipment) of a modern apartment considering proper lighting, power outlets, ventilation, heating and cooling facilities and communication networks giving due attention to safety and health issues.
- Alternate design options should be explored and justification for the selection should be provided. While provisions for all facilities (HVAC, washer/dryer, electric cooking, etc.) should be kept, the total load should not exceed the connected load given by the utility. Design should promote sustainability through the use of energy efficient and low power equipment. The design should propose equipment for sustainability where applicable. Lower cost is another desired characteristics of the design.
- The design must satisfy the requirements set by the latest version of Bangladesh National Building Code (BNBC 2015, Vol. 1, Part 8, Chapter 1). The students are to submit design report at the end of the assignment.
- **Courtesy: Prof. Anisul Haque and Dr. Halima Begum**

Which WPs Are Addressed? In-class Assignment

CEP Attributes

P1: Depth of knowledge
P2: Conflicting requirements
P3: Depth of analysis
P4: Familiarity of issues
P5: Extent of applicable codes
P6: Stakeholder needs
P7: Interdependence

- **WP1:** Power demand requires knowledge of K3 and K8; design requires K5 and K8.
- **WP2:** Requirements are conflicting; all conveniences/facilities are desired while power consumption should be minimized.
- **WP3:** No unique solution; multiple designs are possible. Analysis is needed to select the most appropriate one.

PI of rubric for assessment of CO1, CO2, CO5	WP
Understands objectives, requirements, constraints, standards and codes clearly (CO2)	P1, P2
Analyzes power demand considering requirements and constraints (CO1)	P1, P2
Considers multiple solutions and selects suitable one based on analysis (CO2)	P1, P3
Develops a final design satisfying requirements, constraints, standards and codes (CO2)	P1, P3
Considers safety and health issues in design (CO5)	P1, P2, P3

Which As Are Addressed? In-class Assignment

Activity attributes

A1: Range of resources
 A2: Level of interaction
 A3: Innovation
 A4: Consequences for society and environment
 A5: Familiarity

- **A1:** The design uses resources such as money, information, and technology.
- **A3:** Sustainability and reducing cost require innovative engineering principles and knowledge.
- **A4:** A successful design affects society and environment through sustainability.

Performance Indicator of rubric for assessment of CO4

EA

Explains the design process including engagement of resources clearly

A1

Presents the final design clearly with supporting information and documents

-

Discusses how innovation has been used in design and to resolve conflicting requirements including sustainability

A3, A4

Prepares a report free from mechanical errors (format, grammar, spelling)

-

Example 2: An Open-ended Design Lab

- In this open-ended design lab, students as a group will investigate and design a simple Bangla Speech Recognition system.
- Such a system can be incorporated with a wheelchair to make it voice-controlled and thus enables a physically handicapped person to move freely without the help of a constant care-giver. Moreover, the ability to give commands in Bangla will make the system more user-friendly in the context of Bangladesh.
- The system should be speaker-independent, i.e. it is required that the Bangla speech recognition system should work satisfactorily irrespective of sex, age-group, or dialect of the speaker.
- Students are required to explore different methodologies to investigate the problem through design of experiment and data analysis and select or develop an optimal methodology for design of the system.
- **Courtesy: Prof. Anisul Haque and Dr. Halima Begum**

Which Ps Are Addressed? Open-ended Design Lab

PI of the rubric for assessment of CO5 and CO6 through Lab-report

Performance Indicator	WP
Methodically investigates different algorithms and/or methodologies and organizes evidence/data to demonstrate patterns and highlight differences and/or similarities (CO5)	P1 P3
Selects an optimal algorithm and/or methodology with proper justification (CO5)	P3 P5
Designs a Bangla speech recognition system to meet the requirements subject to the constraints (CO6)	P1 P5
Identifies limitations and implications of the proposed solution (CO6)	P1

- P1:** Requires knowledge of DSP (K4), design of speech processing system (K5), use of DSP tools (K6) and engagement in research literature (K8).
- P3:** Numerous algorithms can be adopted. Choice of the selected algorithm requires in-detail analysis.
- P5:** Requires going beyond the use of standardized features of common English audio commands to design speech recognition application in Bangla.

Example: A Project in the FYDP Course

- **Motivation:** In Bangladesh, depression is found in 2.6–5.5% among men and 6.0–11.8% among women, which causes the increased suicide rate. Early notification of depression can reduce the disease burden.
- **Objective:** A machine-learning-based model will be developed to predict the early depression of the patients from their activities in social media.
- The model will be trained first using data collected from random users.
- A web- or app-based front end will be developed to use the model.
- **Critical challenges:** Various social media data will not be used, as Facebook is the mainstream social media in Bangladesh. Willingness of users to participate, while correctness of the model depends on availability of a large data set. Maintaining privacy of the users' data.
- **Conflicting requirements:** Develop a practical machine-learning model with proper regularization and low variance while limited social media data—only from Facebook—will be used.
- **Courtesy:** Prof. Salekul Islam

Addressing CEP and CEA through the FYDP

- **P1:** Project requires study of existing models with similar goals (K8), data collection from social media (K3, K4), knowledge of design of machine-learning based model (K3, K4), web-based front end (K6) and integration of different components (K5, K6).
- **P2:** Conflicting technical requirements: machine learning model with proper regularization and low variance while limited social media data will be available.
- **P3:** No obvious formulation as a machine-learning problem due to the availability and variations of social media data. Depth of analysis needed to select a specific algorithm from many alternatives.
- **P4:** Computer science and engineering graduates are not typically exposed to issues related to mental health or depression.
- **P7:** Project involves a number of interdependent sub-systems (components), such as, data collection, training module, detection module, front-end application development, etc.
- **A1:** The project must engage diverse resources including people, money, information, and technologies.
- **A2:** Good interaction is required among students, health professionals, and participants.
- **A3:** Innovation is needed to develop the machine-learning model using available Bangla data.
- **A5:** The project deals with a new area for computer science and engineering graduates.

Example: Complex Problem Solving

- Two villages in Timbuktu are separated by a valley about 30 m deep at the deepest section.
- The valley is dry all year except four months, when torrential rainfall can flood major parts of the valley to a depth over 12 m.
- The soil is lateritic with firm bedrock underneath. A bridge connecting the villages is in despair and has to be replaced.
- Write a project brief on how to approach the design for replacement bridge.
- You are limited to locally available building materials; heavy equipment is not available.
- Courtesy: Prof. Megat Johari Mogat Mohd Noor

Aspects to Consider

Economics

Social

Environment

Ethics

Management

Technology

Analysis

Evaluation

Complex problem solving requires balancing multiple non-technical and technical aspects.

Thinking for Replacement Bridge Design

- Site condition
- Weather
- Available technology
- Building materials
- Design
- Costing
- Scheduling

Decision focus

A feasible solution must be technically sound, locally implementable, socially acceptable, environmentally aware, and economically realistic.

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Summary

- Complex Engineering Problem (CEP) solving and Complex Engineering Activities (CEA) are at the core of modern engineering education.
- CEP and CEA differentiate engineering education from education for technologists and technicians.
- CEP = WP1 + at least two from WP2 to WP7.
- CEA = at least two from A1 to A5.
- CEP and CEA can be implemented in theory courses, laboratory courses, design projects, and final year design project.
- Implementation needs careful planning and execution.

Warning

If CEP and CEA are not addressed appropriately in a program, it may produce technologists and technicians — not ‘ENGINEERS’

End of Module

Module 5: Complex Engineering Problems and Complex Engineering Activities



Thank You